Linear and Angular Velocity

Linear and Angular Velocity

Linear velocity is the change in distance, s, divided by the amount of time elapsed, t:

$$v = s/t$$

- If an object is traveling along a circular arc, the distance traveled is the arc length, $s = r\theta$.
- So, we have **linear velocity** (along the arc) as

$$v = \frac{s}{t} = \frac{r\theta}{t} = r\frac{\theta}{t}$$

- Note that $\frac{\theta}{t}$ is the amount of angle change in time t.
- $\omega = \frac{\theta}{t}$ is called the **angular velocity**.

Linear and Angular Velocity

$$v = \frac{s}{t} = \frac{r\theta}{t} = r\frac{\theta}{t} = r\,\omega$$

- Linear velocity is angular velocity multiplied by r for an object traveling around a circle of radius r.
  - ω is typically in radians per unit of time (but can be converted to degrees per unit of time)
  - In practice, angular velocity may be stated in another form, such as rpm, but must be converted to radians per unit of time to use the formula.
  - If angular velocity is given in rpm, we can find linear velocity by converting units and using the circumference for one rotation (examples follow).

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Example:

- A person on a hang glider is moving in a horizontal circular arc of radius 90.0 m with an angular velocity of 0.125 rad/s. What is the person's linear velocity?
- Solution:

$v = r\,\omega$

v = (90.0 m/rad)(0.125 rad/s) = 11.3 m/s. (40.5 km/h)

- (Note that there are 90.0 m/rad on the arc.)

Examples

- The paddles of a riverboat have a radius of 8.50 ft and turn at 20.0 revolutions (or rotations) per min (rpm). What is the speed of a tip of one of the paddles?

Solution: in your notes



$$v = r\frac{\theta}{t} = 8.5\text{ft}\,\frac{20(2\pi)}{\text{min}} = \frac{1068.1\text{ ft}}{\text{min}}$$

$$\frac{2\pi(8.5)\text{ft}}{1\text{ rotation}} \times \frac{20\text{ rotations}}{1\text{ min}} = \frac{1068.1\text{ ft}}{\text{min}}$$

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Examples

- After the brake was applied, a bicycle wheel went through 1.60 rotations before stopping.

  Through how many radians did a spoke rotate?

  Solution:

$$1.6 \text{ rotations} \cdot \frac{2\pi \text{ rad}}{1 \text{ rotation}} = 3.2\pi = 10.1 \text{ rad}$$

- Through how many radians does the minute hand of a clock move in 25 minutes?

  Solution:

$$\frac{25}{60} \text{rotations} \cdot \frac{2\pi \text{ rad}}{1 \text{ rotation}} = \frac{5\pi}{6} = 2.6 \text{ rad}$$

Examples

- A *mil* is a unit of angle measurement used in artillery defined as a central angle of a circle that intercepts an arc equal in length to $1/6400$ of the circumference.
- How many mils are in a central angle of $22.4°$.

Solution:

Convert from degrees to mils by multiplying by mil and dividing by degrees.

$$22.4° \left(\frac{6400 \text{ mil}}{360°} \right) = 398.2 \text{ mil}$$

Linear and Angular Velocity

$$v = \frac{s}{t} = \frac{r\theta}{t} = r\frac{\theta}{t} = r\omega$$

Example: Control-line Model Aircraft

- A model aircraft on a 17.69 m line completes 9 laps in 10.74 sec. What is the linear speed of the aircraft?
- Solution:

$$r\omega = 17.69\text{m}\,\frac{9\text{ rev}}{10.74\text{ sec}} \cdot \frac{2\pi\text{ rad}}{1\text{ rev}}$$

$$= 17.69\text{m}\,\frac{18\pi\text{ rad}}{10.74\text{ sec}} = 93.14\text{ m/sec}$$



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$$= 17.69\text{m}\,\frac{18\pi\text{ rad}}{10.74\text{ sec}} \cdot \frac{3600\text{ sec}}{1\text{ hr}}$$

$$= \frac{335{,}311\text{ m}}{\text{hr}} = 335.3\text{ km/hr}$$

$$= 208.4\text{ mi/hr}$$



Examples

- An interstate route exit is a circular arc 425 m long with a central angle of 75.4°.
  What is the radius of arc of the exit?

$$s = r\theta$$
$$425\text{ m} = r\left(75.4°\,\frac{\pi}{180°}\right)$$
$$r = 323.0\text{ m}$$

Or,

$$s = \frac{\theta}{360°}(2\pi r)$$
$$425\text{ m} = \frac{75.4°}{360°}(2\pi r)$$
$$r = 323.0\text{ m}$$

Solution: in your notes

Examples

- The sweep second hand of a watch is 11.0 mm long. What is the linear velocity of the tip?

$$v = r\frac{\theta}{t} = 11.0\text{ mm}\left(\frac{2\pi}{1\text{min}}\right) = 22\pi\frac{\text{mm}}{\text{min}} \approx 69.1\frac{\text{mm}}{\text{min}}$$

- A vinyl LP record has a diameter of 12 in. and rotates at $33\frac{1}{3}$ rpm.

  What is the linear velocity of a point on the outer edge?

$$v = (6\text{ in})\left(\frac{100}{3}\frac{\text{rot}}{\text{min}}\cdot\frac{2\pi\text{ rad}}{1\text{ rot}}\right) = 400\pi\frac{\text{in}}{\text{min}} \approx 1256.6\frac{\text{in}}{\text{min}}$$

or $$v = \frac{100}{3}\frac{\text{rot}}{\text{min}}\cdot 2\pi(6)\frac{\text{in}}{\text{rot}} = 400\pi\frac{\text{in}}{\text{min}}\cdot\frac{1\text{ft}}{12\text{in}}\cdot\frac{60\text{ min}}{1\text{ hr}} = 2000\pi\frac{\text{ft}}{\text{hr}}$$

Solution: in your notes

Examples

- The flywheel of an engine is 0.36 m in diameter.
  If it is rotating at 750 rot/min (rpm), through what distance does a point on the rim move in 2.00 s?

$$v = 0.18\text{ m} \cdot \frac{750\text{ rot}}{\text{min}} \cdot \frac{2\pi}{\text{rot}} = 848.2\frac{\text{m}}{\text{min}} \cdot \frac{1\text{ min}}{60\text{ sec}} = 14.137\frac{\text{m}}{\text{sec}}$$

So it moved $2(14.137) = 28.27$ m

- A drill bit $\frac{1}{2}$ in. in diameter rotates at 1800 rpm.
  What is the linear velocity of a point on its circumference?

$$v = \frac{1}{4}\text{in}\frac{1800\text{ rot}}{\text{min}} \cdot \frac{2\pi}{\text{rot}} = 900\pi\frac{\text{in}}{\text{min}} = 2827\frac{\text{in}}{\text{min}}$$

Solution: in your notes

Examples

- Through what total angle does the drive shaft of a car rotate in 1 s when the tachometer reads 3200 rpm?

Solution: in your notes

Examples

The blades on the GE Model 1.5s, 1.5 Mw wind turbine (windmill) are 116 ft long.

- What is the linear velocity the tip of the blade, if the windmill rotates at 22.0 r/min?
- What is the surface area of the rotating blades measured in acres? (1 acre = 43,560 ft^2)

Solution: in your notes

- Wind Fast Fact #17: As of May 2016, the United States is home to 48,500 operating wind turbines. These turbines are present across 40 states plus Puerto Rico. - http://windenergyfoundation.org/
- Common Industrial Wind Turbines: http://www.aweo.org/windmodels.html
- http://www.awea.org/Issues/Content.aspx?ItemNumber=832&gclid=CjwKEAjw8ZzHBRCUwrrV59XinXUSJADSTE5kPm_mWpkB7_JfzrycKdFeDjg7aQ9J3BV5weUCtXgxHBoCFu7w_wcB
- http://savetheeaglesinternational.org/new/us-windfarms-kill-10-20-times-more-than-previously-thought.html